

1-C:

This condition is derived from the principle of **equi-marginal principle** which states that a consumer maximises total utility when the **marginal utility per unit of currency is equal for all goods**.

That condition is:

$$\frac{MU_Y}{P_Y} = \frac{MU_Z}{P_Z}$$

Rearranging this gives:

$$MU_Y \times P_Z = MU_Z \times P_Y$$

So, option C correctly expresses this equilibrium condition. It ensures that **reallocating expenditure** between Y and Z would not increase total utility, i.e., the consumer is at equilibrium.

Here is why the other options are incorrect:

A: A says $MU_Y = MU_Z$. Equal marginal utilities alone don't ensure utility maximisation.

B: B gives $MU_Y \times P_Y = MU_Z \times P_Z$, which doesn't follow from any utility maximisation condition. In fact, it multiplies marginal utility with its own price, which doesn't help us compare utility **per dollar** spent on both goods.

D: D claims we can't tell from the information given but we actually have everything needed: **marginal utility and price**. That's sufficient for applying the equi-marginal principle.

2-B:

B) This is the **only incorrect statement** about indifference curves. **Indifference curves can never intersect**. If two curves intersect, it would mean a consumer is **indifferent between two different levels of utility**, which is logically impossible.

Here is why the other options are incorrect:

A: Indifference curves are **usually convex to the origin** due to the **diminishing marginal rate of substitution (MRS)** — consumers are willing to give up less of one good to get more of another as they consume more of it.

C: Consumers always prefer higher indifference curves because they represent **more utility**.

D: The **slope of the indifference curve is the MRS**, which reflects the rate at which a consumer is willing to trade one good for another while maintaining the same level of satisfaction.

3-A:

A – when the cost of producing the last extra unit equals the value the consumers place on it.

This is the definition of **allocative efficiency**. Allocative efficiency occurs where **price = marginal cost ($P = MC$)**, because the **price reflects the consumer's valuation**, and MC reflects the **cost to society** of producing the last unit.

Here is why the other options are incorrect:

B: Minimising marginal cost does not help to achieve allocative efficiency.

C: Even if total cost equals total value to consumers, it doesn't mean we're at **the socially optimal output level**. $TC=TR/TE$ represents the normal profit condition.

D: Maximising total revenue has nothing to do with allocative efficiency.

4-C:

To find **supernormal profit**, we use:

Supernormal profit = Total Revenue – Total Cost

SNP occurs when $TR > TC$

Profit maximisation occurs where **$MC = MR$** . From the table:

- At output 3, **$MR = 45$** and **$MC = 45$** → this is the profit-maximising output.
- At output 3:
 - $AR = 85 \rightarrow \text{Total Revenue (TR)} = 3 \text{ (qty)} \times 85(AR) = \mathbf{255}$
 - $ATC = 75 \rightarrow \text{Total Cost (TC)} = 3 \text{ (qty)} \times 75 (ATC) = \mathbf{225}$
 - Supernormal Profit = $255 - 225 = \mathbf{30}$

5-D:

Firms in monopolistic competition operate at the profit maximizing level of output where $MC=MR$.

In the **long-run equilibrium of monopolistic competition**, firms make **normal profit ($AR = AC$)**, but still produce **less than the productively efficient level** (i.e., not at minimum AC). In **Diagram D**, $AR = AC$ at Q_e , which reflects **zero economic profit**, as expected in long-run equilibrium due to **free entry and exit**. However, the output is still **below the productively efficient level**, hence inefficiency remains. This explains why key D is correct.

A: In Diagram A, AR lies **above AC**, which implies **supernormal profit**. That's not sustainable in the **long run** of monopolistic competition due to market entry.

B: In Diagram B, AR lies **below AC**, indicating a **loss** — again, not sustainable in long-run equilibrium. Also, we are not operating at the profit maximizing level of output where $MC=MR$.

C: Again, in diagram C, we are not operating at the profit max level of output where $MC=MR$. Here also, $AR > ATC$, indicating supernormal profits.

6-A:

In a perfectly contestable market, **entry and exit are costless**, and firms can freely enter when profit is available and exit without incurring losses. However, in real-world markets, **sunk costs** like non-recoverable investments in capital or marketing prevent this. Hence, the key issue is **irrecoverable fixed costs** — even if a firm exits, it loses the money already spent, making markets less than perfectly contestable. This explains why key A is correct.

B: This is about output levels and equilibrium, not contestability. It doesn't explain barriers to entry or exit.

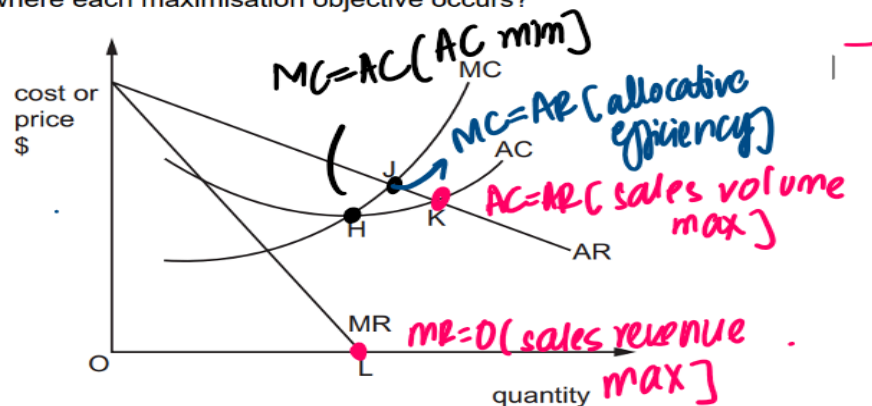
C: Large firms may grow **without expanding capital**. But it does not explain why some firms in contestable markets are not perfectly contestable i.e. they do not have cost free entry and exit.

D: Multinational companies may benefit economies of scale (but not always, they may experience diseconomies of scale as well). Regardless, it does not explain why some firms in contestable markets are not perfectly contestable, i.e., they do not have cost free entry and exit.

7-B: On a firm's revenue diagram, **revenue maximisation** occurs where $MR = 0$. Point “L” lies where $MR = 0$, showing total revenue is at its highest.

Sales maximisation occurs where $AR = AC$, i.e., normal profits are made while selling the highest possible quantity. This is at point “K”. So B correctly identifies both points.

Which points show where each maximisation objective occurs?



8-D:

External economies of scale are cost reductions that benefit all firms in an industry due to **industry-wide growth**, not actions of an individual firm. A local college offering relevant training improves the overall quality of the labour force for **all firms**, thus reducing average costs across the sector, making key D correct.

A: Purchasing inputs in bulk is an **internal economy of scale** (purchasing EOS)— achieved by the individual firm due to its own scale.

B: Firm benefiting from lower interest rates due to its size represents an **internal economy of scale** (financial EOS)

C: Increased labour productivity can result from firm-specific training or technology, making it an **internal**, not external, source of scale economies. Additionally, this represents labor economies of scale.

9-A:

A **cartel** is an agreement among producers (usually in an oligopoly) to **fix prices or restrict output** to maximise joint profits, often leading to higher prices and lower output than in competitive markets. The question describes exactly that — an agreement to restrict supply or set prices.

B: A conglomerate merger involves firms in **unrelated industries** (it represents “**diversification**”), not collusion between producers of the same product.

C: Economies of scale refer to cost reductions due to increased production, not market manipulation.

D: Horizontal integration means merging with a **competitor** at the same production stage, but not necessarily forming a cartel or fixing prices.

10-A:

A **four-firm concentration ratio** tells us the **total combined market share** of the four largest firms in the industry. A value of 25% means these four firms collectively control **a quarter of the market**, indicating low concentration. This explains why key A is correct.

11-C:

Allocative efficiency is achieved when **$P(AR) = MC$** , meaning the price consumers are willing to pay (value) equals the cost of producing the last unit. In a monopoly, output is **restricted** and price is usually set above marginal cost. A government can correct this **market failure** by imposing a price cap where the **demand curve (AR) intersects the marginal cost curve** — this is point **C** in the diagram.

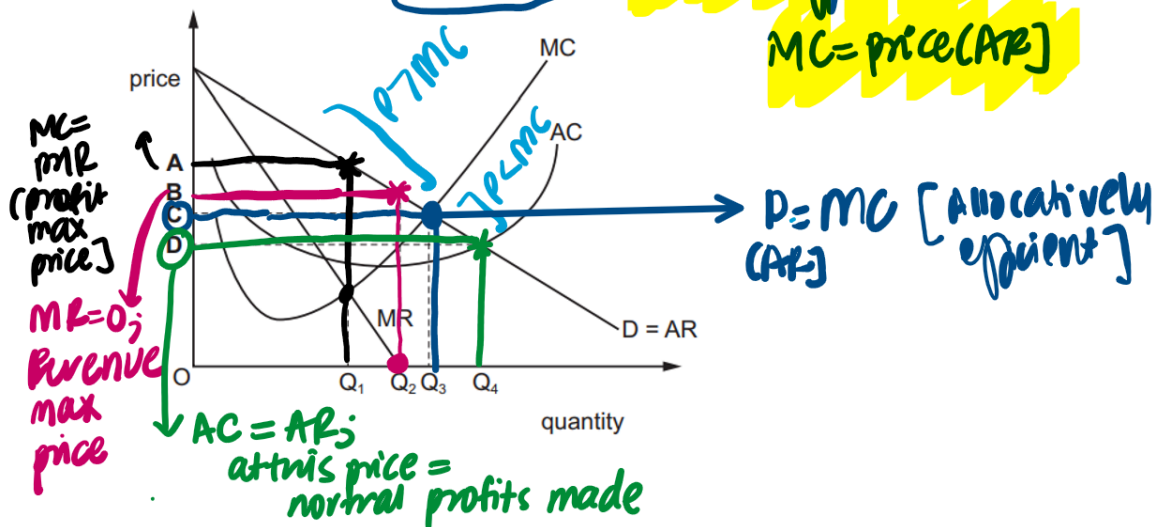
A: This is likely the monopoly price where $MR = MC$ — profit maximising, but **allocatively inefficient** since $price > MC$.

B: The output level corresponding to this price level represents the revenue maximizing quantity. Hence, this price would be set if the firm pursues revenue maximization.

D: This is where $AC=AR$, which depicts a normal profit situation. Here allocative efficiency won't be achieved.

A government decides to use price controls to achieve allocative efficiency in a monopoly market.

Which price would need to be set to achieve allocative efficiency?



12-A:

A **wage increase above the equilibrium level** (W_1 to W_2) leads to a **surplus of labour**, i.e., unemployment. At the higher wage W_2 , qty demanded of labor (which will represent the new level of employment) contracts from L_1 to L_2 , as now fewer workers are demanded (L_2).

So employment falls from L_1 (previous level) to L_2 (new level). This is a classic case of **real wage (classical) unemployment**. Hence making A correct.

B: L_3 isn't the original employment level as the equilibrium level of employment was L_1 where $DL=SL$.

C: Employment can't rise if wages are **above** equilibrium. A rise from L_1 to L_3 would mean a **rise in employment**, which contradicts theory.

D: This again implies **increasing employment**, which again contradicts the **excess supply** condition after a wage rise above equilibrium wage.

13-B: When **marginal productivity of labour rises**, each worker contributes more to output. Since labour is a **derived demand**, this means the value of hiring additional workers increases, leading to an **outward shift in the demand for labour**. Firms are willing to hire more at every wage level. We can also understand it like this:

$$MRP(DL) = MPL \times MR$$

$\uparrow$ (marginal revenue product $\uparrow$) $\rightarrow$ **outward shift**

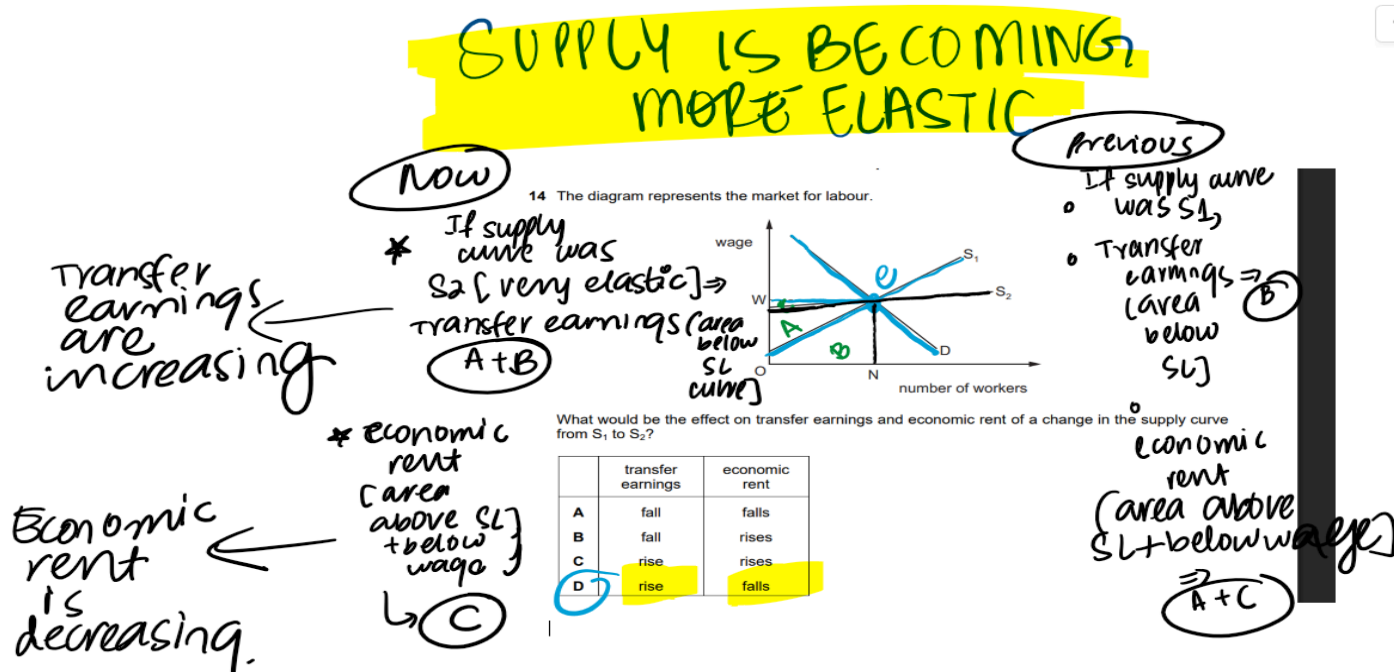
$\uparrow$ (marginal productivity of workforce $\uparrow$)

A: A leftward shift in demand would occur if labor was now **less productive**, which contradicts the effect of better machinery.

C: A shift in **supply** would occur only if something affected workers' **willingness or ability to work**, not their productivity.

D: Same as C

14-D: When the supply of labor curve becomes more elastic, it means that the quantity supplied is more responsive to wage changes. **This leads to an increase in transfer earnings (the minimum amount a factor of production needs to earn to stay in its current use) and a decrease in economic rent (the amount earned above the transfer earnings).**



15-A:

Explanation (Correct: A – decrease in tax-free allowance, increase in means-testing):

A **poverty trap** occurs when low-income individuals face **little incentive to work more** due to loss of benefits and higher taxes.

-> Reducing the **tax-free allowance** means they start paying tax at lower incomes.

-> while increasing **means-testing** means they lose substantial benefits that they could have received when they were unemployed, as income rises .

This combination **intensifies the trap**, leaving workers worse off for earning more. This will put them off from working.

16-D:

A **price comparison website** helps consumers compare the prices of goods; this is a classic example of money serving as a **unit of account**, i.e., a common measure to assign and compare values.

Without this function, **value comparison** between goods would be difficult. The website enables efficient resource allocation by allowing consumers to evaluate the relative worth of goods. Hence making key D correct.

A: Medium of exchange refers to money being used to **facilitate transactions** (not the act of **comparing prices**)

B: Standard of deferred payment means money can be used to **settle debts in the future**, e.g., loan repayment (unrelated to comparing prices today).

C: Store of value means money retains its value over time — again (not applicable to comparison of prices at a given moment)

17-D:

Statement 1 describes **voluntary unemployment** — individuals might choose **not to work** if benefits are attractive and working incurs tax. They opt out of work due to incentives.

Statement 2 describes **involuntary unemployment** — the person is actively seeking work but is unable to find a job due to **economic downturn**, a classic example of **cyclical unemployment**.

A: Statement 1 is not cyclical; it's driven by incentives/disincentives. Structural doesn't describe statement 2 accurately — it's about mismatches in skills of workers and skills required by jobs.

B: Frictional unemployment involves **transitions** between jobs, not deep recessions. Statement 2 clearly reflects **lack of overall demand** in the economy, not friction.

C: Structural unemployment involves long-term shifts in the economy, making the explanation for statement 1 incorrect.

18-A:

In a **recession**, even if interest rates are cut to encourage borrowing and investment, the private sector might not respond if **confidence is low**. This is known as a **liquidity trap** or **Keynesian ineffectiveness of monetary policy**: firms delay investment and consumers delay spending due to pessimism about future economic prospects. So even cheap credit won't boost aggregate demand.

B: Interest rate cuts don't **cause deflation**; they are usually used to **combat** it.

C: Lower interest rates tend to **depreciate** a currency (not appreciate it).

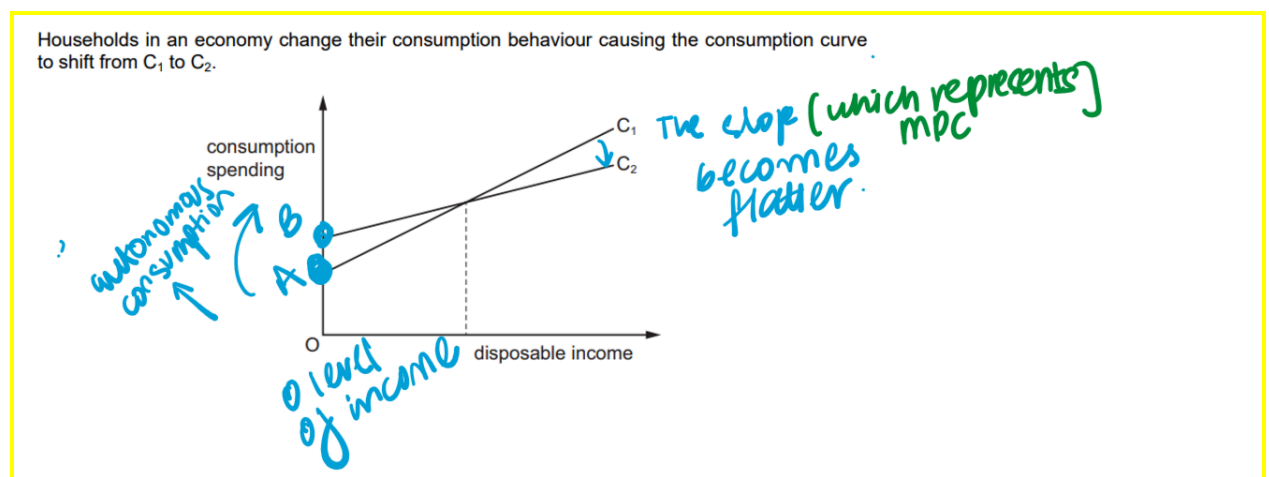
D: An interest rate cut **does not lead to a decrease in money supply**. A higher money supply usually corresponds to a lower interest rate.

19-C:

A shift in the consumption function from C_1 to C_2 **upwards and flatter** means:

- **Autonomous consumption** i.e. **consumption independent of changes in consumers' income** (the y-intercept) has increased — people are spending more **even when income is zero**.
- The **marginal propensity to consume (MPC)** has decreased; this is reflected in a flatter slope of the consumption function, meaning people are spending a lesser proportion of their additional income earned.

Note: The gradient of the consumption function (or its slope) represents the Marginal Propensity to consume.



20-A:

Lower interest rates reduce the **costs of borrowing**, making it more attractive for businesses to invest and expand. This leads to **higher demand for labour**, more output, and hence **lower demand-deficient unemployment**. Hence key A will be correct.

B: A decrease in the budget deficit represents a contractionary fiscal policy, which can exacerbate unemployment.

C: Interest rate cuts cause **depreciation** in currency (due to capital outflows), not an increase in the exchange rate.

D: Lower interest rates reduce the **opportunity cost of spending**, not increase it.

21-C:

This is a reference to the **short-run Phillips curve**, which shows an inverse relationship between inflation and unemployment.

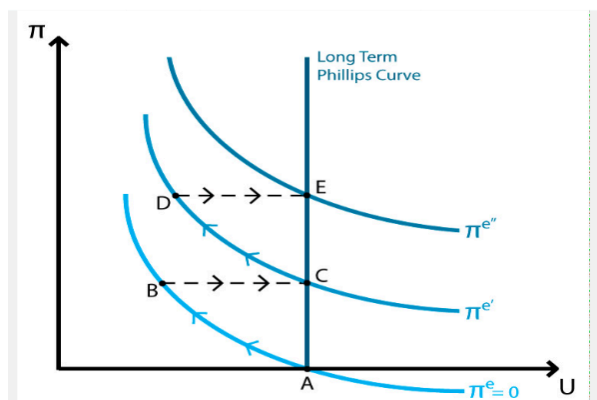
-> However, in the **long run**, expectations adjust. If inflation rises, workers demand higher wages, pushing **costs up (SRPC shifts to the right as Aggregate supply reduces)**, and unemployment may return to its **natural rate**.

-> This process is known as **expectations-augmented Phillips curve theory**, explaining why the trade-off holds only in the short run. Hence making key C correct.

A: This describes the **short-run** trade-off but doesn't explain **why** the curve is only short-run — expectations are the key.

B: Suggests high inflation **causes unemployment**, which is not the logic behind Phillips. The Phillips curve would show how low unemployment corresponds to a higher level of inflation. In any case, it does not explain why the trade-off holds only in the short run.

D: Inflation lowering real wages doesn't explain the **temporal limit** of the short-run trade-off.



22-D:

Negative **actual growth** means a fall in real GDP. Lower output leads to **lower derived demand for labour**, so **unemployment rises**. It's a natural outcome of economic downturns.

A: The financial account is unlikely to be directly affected — surplus or deficit depends on capital flows, not real GDP. However, negative growth may discourage foreign capital flows by investors, which means that we can expect a decrease in surplus in the financial account.

B: A current account deficit might actually **decrease** in a downturn, as falling incomes reduce **imports**.

C: Demand-pull inflation generally **falls** in recessions due to weaker Aggregate demand.

23-B:

Monetary policy is more effective when the sectors it targets are **responsive** to interest rate changes. If investment by firms is **interest elastic**, a cut in interest rates will **significantly increase investment**, boosting aggregate demand and employment.

A: **Income elasticity** of consumption is less relevant here ; the policy tool is **interest rate**, so we are interested in “interest elasticity.”

C: High leakage (via saving, imports, or taxation) weakens the multiplier, reducing the policy's effectiveness.

D: A reduction in the interest rate leads to depreciation of the exchange rate, which will make export-oriented goods relatively cheaper. This will lead to the qty demand of export oriented goods to extend by a smaller percentage if their **demand is price inelastic**, leading to a fall in export revenues. This will reduce the AD of the economy.

24-C: When a government wants a **rapid domestic impact**, it's more concerned with **domestic demand, investment, and employment**.

Therefore, the time it takes for there to be effects on the **foreign exchange rate and the international market** will not be that of a concern for the government.

Since the foreign **exchange rate lag is least relevant** to a domestically focused policy, key C becomes correct.

A: Recognition lag can delay any policy response entirely ; it's a significant concern.

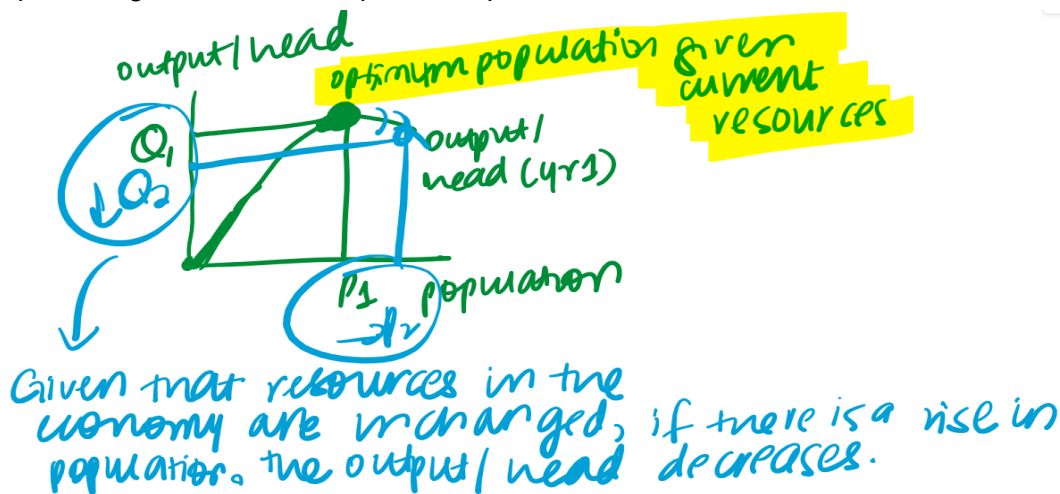
B: Response lag — the time the **domestic economy takes to react** — is a major constraint.

D: Implementation lag (enacting the policy) can also delay domestic impact, which the government is concerned about.

25-A:

The diagram shows **output per head has increased** at every population level — the curve has **shifted up**.

However, **an increase in population alone would cause a movement along the curve**, not a shift. A rightward movement could even reduce output per head (due to diminishing returns). So population growth **cannot** explain an upward shift in the curve. This makes A correct.



B: Improved technology raises **productivity**, shifting the output per head curve **upward** — valid explanation.

C: More capital means **resources of the economy are increasing, which is a valid cause of the shift**.

D: An increase in the volume of productive resources in use would again bring a shift in the output per head curve. This would lead to output per head to increase at every population level.

26-D:

Raising income tax (contractionary fiscal policy) lowers **disposable income**, causing **consumers to cut spending**, including on imports. This method is commonly used to address **current account deficits**. This is an expenditure reducing policy.

A: Devaluation is an **expenditure-switching** policy; it shifts spending from foreign goods to domestically produced goods by making imports expensive for domestic residents.

B: A subsidy on domestic production would also lead to **expenditure switching** from imported goods to local goods, as domestic goods are now cheaper.

C: A tariff is also an expenditure-switching policy; it shifts spending from foreign goods to domestically produced goods.

27-A:

To calculate the change in **real income per head**, apply this formula:

$$\text{Real income growth} = \text{Nominal income growth} - \text{Inflation}$$

$$\text{Real income per capita growth} = \text{Real income growth} - \text{Population growth}$$

Given:

Nominal income $\uparrow$ 5%

Inflation $\uparrow$ 4% $\rightarrow$ Real income $\uparrow$ 1%

Population $\uparrow$ 2% $\rightarrow$ Real income per capita = 1% - 2% = -1%

Since real income per head decreases by 1%, key A becomes correct.

28-A:

A) FDI (in the form of MNCS) would lead to an increased output being produced in the host country, and when this also increases the volume of exports, it can improve the trade balance of the developing economy, making option A correct.

B) FDI **adds to the stock of capital**, but much of the **investment income generated (profits)** is **repatriated back to the foreign investor's home country**. This makes this incorrect.

C) FDI typically **increases** tax revenues, as new firms register, workers earn incomes, and governments collect corporate taxes.

D) FDI creates **more demand for labour** (especially in manufacturing and services), which leads to **higher wages**, not lower.

29-A:

The Gini coefficient is calculated by taking the area between the Lorenz curve and the line of perfect equality (**AREA Y**), and dividing that area by the total area under the line of perfect equality (**AREA Y+Z**).

The **Gini coefficient** measures income inequality based on the **Lorenz Curve**:

- Area Y = between Lorenz curve and 45° line (perfect equality)
- Area Z = below the Lorenz curve
- **Gini = Area Y / (Y + Z)**

30-A:

High-income countries have **large tertiary sectors**, small primary sectors, and **well-developed manufacturing**.

- Country A has the **highest tertiary share (45%)**, and **lowest primary (15%)**, typical of a developed, service-driven economy.
- This structure reflects that employment is in services like healthcare, finance, and education.

